

Quantum Feature Extractor for Multi-Class Brain Tumor Classification

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April 2026

1 Introduction

The landscape of computational oncology is currently undergoing a radical transformation as the limitations of classical deep learning become increasingly apparent in the face of escalating data complexity. Traditional Convolutional Neural Networks (CNNs), while highly effective for image processing, frequently encounter performance saturation problem when tasked with identifying subtle pathological variations in high-dimensional medical images. As the depth of these networks increases, the risk of vanishing gradients and the massive computational overhead required for processing high-resolution medical images pose significant barriers to real-time clinical deployment. Quantum Machine Learning offers a potential resolution to these challenges by leveraging the fundamental principles of quantum mechanics—superposition and entanglement—to process information in a fundamentally different manner.

Quantum computing operates through qubits, which unlike classical bits, can exist in a superposition of states $|0\rangle$ and $|1\rangle$. This allows a quantum circuit to represent and process multiple data states simultaneously, a phenomenon known as quantum parallelism. In the context of brain tumor classification, where the distinction between malignant gliomas and benign meningiomas often relies on intricate textures and vascular patterns, the ability of quantum states to inhabit a high-dimensional Hilbert space provides a significant representational advantage.

This report presents a novel integration of the QuFeX module [8] into a Y-Net architecture [3] for the classification of brain tumors from MRI data. By situating the QuFeX module at the architectural bottleneck, the model benefits from the efficient hierarchical feature extraction of classical CNNs and the superior expressive power of quantum circuits. The primary contributions of this work include the development of a hybrid quantum-classical feature extraction pipeline that circumvents the dimensionality issues typical of near-term quantum devices, and the demonstration of enhanced classification accuracy for quaternary tumor detection.

2 Related Works

Quantum neural networks (QNNs) incorporate parameterized quantum circuits (PQCs) as layers of a neural model. Cong et al. first proposed the QCNN architecture [1], mimicking convolution and pooling in a quantum circuit. Other works have designed variational quantum classifiers, e.g. Farhi and Neven’s quantum circuit model for image recognition [2] and Havlíček et al quantum kernel classifier [4]. There have been attempts to improve classification metrics on classical data using quantum circuits. Some works, like [7] focussed on binary classification propose scaling down input data by using traditional dimensionality reduction techniques like Principal Component Analysis to reduce the number of features. The success of QCNNs has been widely attributed to the property of global feature entanglement by quantum filters when compared to classical CNNs that focus more on locality-based computation. Multiclass classification paradigm has also been dealt with in the quantum method, highlighting faster convergence with fewer parameters and comparable accuracies to classical networks. Methods like [10] propose that QCNN outperform classical CNNs when the number of classes increase. These approaches showed that small quantum circuits can capture complex feature maps for classification, although scalability remains a challenge.

Quanvolutional networks were introduced by Henderson [6] as a hybrid scheme where classical convolutional layers are augmented by fixed (or optionally trainable) quantum circuits on local image patches. In practice, each image patch is embedded into a quantum circuit, and measurements produce enhanced feature maps, which are then fed to classical layers [6, 9]. Quanvolution has been demonstrated to improve classification on datasets like MNIST with relatively shallow circuits [6, 9]. The main advantage is leveraging quantum kernels locally, but a drawback is the need for many parallel quantum evaluations (one per patch).

Quantum ML for classification tasks has been explored in various contexts. Recent reviews and experiments report that hybrid models can achieve strong image classification performance with fewer parameters than classical networks [11]. In medical imaging, quantum approaches are emerging: for example, Hashemzahi et al. propose the Y-Net, a U-Net-based CNN, for MRI tumor classification [3]. Wang et al. introduce HQNet, a hybrid quantum-classical CNN for multi-class brain MRI tumor detection [12]. These indicate that embedding quantum layers can benefit brain imaging tasks. Nevertheless, most prior work either focuses on classification with shallow quantum layers or on segmentation with architectures like U-Net; applying advanced quantum modules like QuFeX to classification remains largely unexplored. Jain and Kalev [8] recently introduced QuFeX for semantic segmentation (replacing the U-Net bottleneck), achieving better feature extraction. We build on this by adapting QuFeX to tumor classification.

3 Methodology

3.1 Quantum Neural Networks vs. Quanvolution

Two paradigms guide our design. In QNNs, a trainable quantum circuit (often a variational circuit) acts as one or more layers of the network. For example, a PQC can replace fully-connected layers at the end of a CNN or serve as an entire classifier [2]. Such circuits can capture global patterns but may require large state spaces. In contrast, quanvolutional networks place small quantum circuits on local regions: each patch of the input (or of a feature map) is independently processed by a fixed or random quantum transformation, and the resulting expectation values form a new feature map [6]. Quanvolution thus injects quantum nonlinearity locally, akin to a convolutional filter, but at the cost of many circuit evaluations (one per patch).

QuFeX merges these approaches. As proposed by [8], QuFeX uses trainable quantum circuits on multiple feature-map channels simultaneously, compressing information like a QCNN while preserving the spatial locality of quanvolution. Specifically, given a set of classical feature maps (from convolutional layers), QuFeX extracts small patches across channels and feeds each patch into a parameterized quantum circuit. The circuit’s qubits encode the patch values and apply an ansatz of trainable rotations and entanglers. After measuring Pauli-Z expectations of the qubits, the results are aggregated to form output feature maps. Thus each QuFeX layer transforms an input tensor into a reduced set of output channels, akin to a dimensionality-reducing convolutional bottleneck. The hybrid of QCNN and Quanvolution in QuFeX enables extracting rich features with relatively few quantum circuits [8]. We adopt this philosophy: QuFeX acts as a trainable quantum “bottleneck” that complements the classical CNN pipeline.

3.2 Adapted Hybrid Architecture

We design a hybrid quantum-classical CNN for multi-class MRI tumor classification visualized in 1, details for the same are as follows:

- **Classical Encoder:** We use a convolutional backbone to extract hierarchical features from the input MRI. Options include a U-Net-style encoder or a ResNet-like feature extractor. A typical choice is successive blocks of convolution (with ReLU) and pooling. For example, the encoder might consist of 3–4 stages with filter sizes increasing (e.g. 32, 64, 128, 256) and spatial size halving at each down-sample. This yields a high-level feature tensor (the “bottleneck”) of modest spatial size and many channels (e.g. 128 or 256).
- **Classical Decoder:** A similar convolutional backbone to that of the encoder is used in order to extract hierarchical features from the feature maps. A typical choice is successive blocks of convolution (with ReLU)

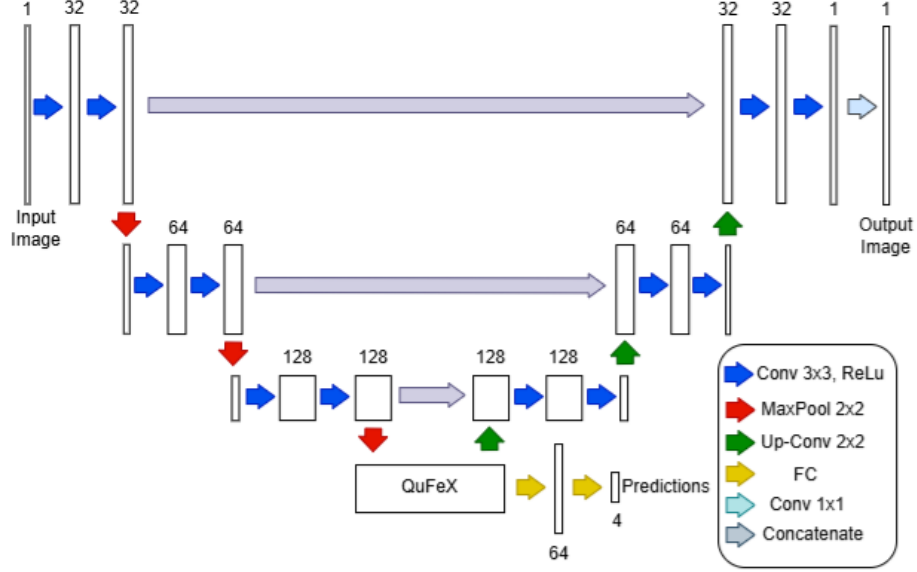


Figure 1: Y-Net Architecture

and upsampling. For example, the decoder might consist of 3–4 stages with filter sizes increasing (e.g. 32, 64, 128, 256) and spatial size doubling at each up-sample. This yields a reconstructed image of same spatial size and channels as the input image.

- **Quantum Bottleneck (QuFeX):** At this point we replace or augment the classical bottleneck with QuFeX. The encoded feature tensor is divided into small patches. Each patch is fed into a parameterized quantum circuit. In practice, n qubits can be used to encode an n -dimensional patch. The circuit applies rotations on each qubit with angles proportional to the patch’s pixel intensities, followed by layers of entangling gates and trainable rotation gates. We then measure each qubit in the Z-basis. The measured values (or their average) form the elements of the output feature maps. Multiple circuits (on disjoint patches) can run in parallel with shared parameters, producing a transformed tensor of reduced size.
- **Residual / Skip Connections:** Inspired by ResNet [5] and the original Qu-Net design [8], we include identity skip connections around the QuFeX module. Concretely, we add the input tensor (from the encoder) to the output of the QuFeX layer (after appropriate reshaping or projection) before passing on to the next stage. This allows the model to bypass the quantum transformation if it does not help, stabilizing training and ensuring the hybrid model can fall back on classical features if needed [8].

If using a U-Net-style encoder, we also retain the usual encoder-decoder skip links (in a Y-Net variant) to combine low- and high-level features.

- **Classifier Head:** For classification, we collapse the final feature maps into class scores. A common choice is global average pooling followed by one or two fully connected (dense) layers and a softmax. For instance, after QuFeX we may apply a 1×1 convolution to reduce channel count, then global-pool and connect to a 3-unit output layer (for the three tumor classes). Alternatively, if using a Y-Net (dual-branch) architecture, the encoder’s bottleneck feeds two heads: one deconvolutional branch for segmentation (if labels are available) and one classification branch (for tumor type). In our single-task setting, we use the simpler single branch. We can also explore adding dropout or batch normalization before the final layer to combat overfitting on limited MRI data. The network is trained with a multi-class cross-entropy loss.

Table 1: YNet Autoencoder - Summary (Batch Size = 16, Input: $1 \times 64 \times 64$)

Module	Layer	Output Shape	Params
UNet Encoder		$16 \times 128 \times 16 \times 16$	–
	DoubleConv (Sequential)	$16 \times 32 \times 64 \times 64$	9,568
	MaxPool2d	$16 \times 32 \times 32 \times 32$	–
	DoubleConv (Sequential)	$16 \times 64 \times 32 \times 32$	55,424
	MaxPool2d	$16 \times 64 \times 16 \times 16$	–
	DoubleConv (Sequential)	$16 \times 128 \times 16 \times 16$	221,440
Bottleneck Conv2d		$16 \times 4 \times 16 \times 16$	516
QuFeX (Quantum Feature Extractor)		16×256	–
	Unfold (kernel 2×2 , stride 2)	$16 \times 16 \times 64$	–
	TorchLayer (4-qubit circuit)	1024×4	4
Classifier Head (Sequential)		16×4	–
	Linear ($256 \rightarrow 64$)	16×64	16,448
	Linear ($64 \rightarrow 4$)	16×4	260
UNet Decoder		$16 \times 1 \times 64 \times 64$	–
	ConvTranspose2d	$16 \times 64 \times 32 \times 32$	32,832
	DoubleConv (Sequential)	$16 \times 64 \times 32 \times 32$	110,720
	ConvTranspose2d	$16 \times 32 \times 64 \times 64$	8,224
	DoubleConv (Sequential)	$16 \times 32 \times 64 \times 64$	27,712
	Conv2d (output)	$16 \times 1 \times 64 \times 64$	33
Total Trainable Parameters		483,181	

QuFeX TorchLayer parameters represent 4 quantum rotation angles.

3.3 Training Strategy

The hybrid model is trained end-to-end in a unified framework where classical layers and quantum circuit parameters are jointly optimized using the Adam optimizer. To achieve high-fidelity classification alongside accurate feature preservation, the objective function is a composite of two distinct loss terms. For the classification head, we employ Categorical Cross-Entropy Loss (L_{CLS}), which measures the divergence between the predicted probability distribution $p_{o,c}$ and the ground truth labels $y_{o,c}$ for all tumor classes M . This function penalizes the model logarithmically, ensuring that the network assigns high confidence to the correct pathological class. Simultaneously, the decoder branch utilizes Mean Squared Error (MSE) Loss (L_{MSE}) as the reconstruction criterion. By calculating the average squared difference between the original input pixels $x_{i,j}$ and the reconstructed pixels $\hat{x}_{i,j}$ across the image dimensions W and H , the model is forced to retain the most salient spatial features of the brain scans. The final loss (L) is a weighted sum of the classification loss (L_{CLS}) and reconstruction loss (L_{MSE}) with α as the weight ($0 \leq \alpha \leq 1$) applied on the reconstruction loss (L_{MSE}) to ensure that the model remains proficient in handling its primary task i.e. classification. Since quantum circuits can be prone to decoherence and noise, we incorporate residual connections to alleviate vanishing gradients and stabilize the optimization landscape. To ensure the model generalizes well to unseen clinical data, we implement a robust data augmentation pipeline—including rotations, flips, grayscale conversion, and resizing.

$$L_{CLS} = - \sum_{c=1}^M y_{o,c} \log(p_{o,c}) \quad (1)$$

$$L_{MSE} = \frac{1}{WH} \sum_{i=1}^W \sum_{j=1}^H (x_{i,j} - \hat{x}_{i,j})^2 \quad (2)$$

$$L = L_{CLS} + \alpha L_{MSE} \quad (3)$$

In summary, our adapted architecture embeds QuFeX at the CNN’s bottleneck, with careful design of filter counts, circuit size, and skip connections to suit classification. We will compare variants to determine the best approach for brain tumor MRI classification.

4 Experiments

We primarily evaluate our proposed method on two distinct architectural configurations, each designed to probe a different aspect of quantum-classical feature integration. The first architecture comprises the encoder branch of a U-Net coupled with the Quantum Feature Extractor (QuFeX) module at its output. This configuration is intentionally kept lightweight — retaining only the contracting path of the U-Net to progressively extract spatial hierarchies — and is optimized solely using Cross Entropy Loss [1]. This allows us to assess the discriminative capacity of QuFeX in isolation, without the confounding influence of a reconstruction objective.

The second architecture employs the complete U-Net with skip connections intact, wherein the bottleneck layer is replaced by the Quantum Feature Extractor (QuFeX) module. The skip connections bridge the encoder and decoder at multiple resolutions, preserving fine-grained spatial information that would otherwise be lost through successive downsampling. In this setting, the network is jointly optimized using both Cross Entropy Loss [1] and Mean Squared Error Loss [2], combined through a weighted objective [3] with $\alpha = 1$, assigning equal contribution to both terms. This dual-loss formulation encourages the model to simultaneously produce accurate class predictions and faithful reconstructions, placing greater demand on the representational capacity of the quantum bottleneck.

4.1 Results

Our primary aim is to compare the performance of the model using the QuFeX module for multiclass classification of Brain Tumor MRI data with other established frameworks. We explore different architectures inspired from the U-Net module. The results are as follows:

Table 2: Results			
Encoder	Decoder	Bottleneck	Accuracy
✓	✗	U-Net	80.64%
✓	✗	QCNN	76.95%
✓	✗	Quanvolution	78.08%
✓	✗	QuFeX	79.11%
✓	✓	U-Net	83.85%
✓	✓	QCNN	77.15%
✓	✓	Quanvolution	78.05%
✓	✓	QuFeX	84.31%

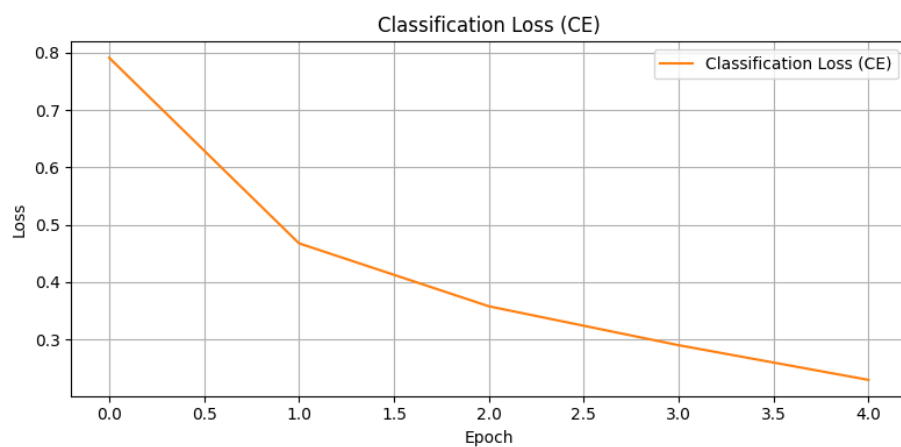


Figure 2: Loss Plot for Encoder only Y-Net

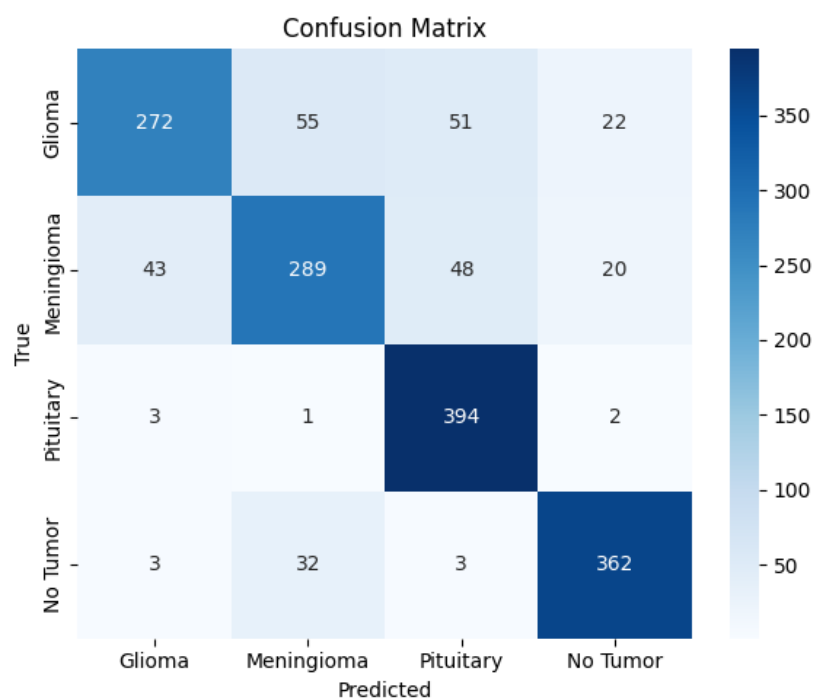


Figure 3: Confusion Matrix for Encoder only Y-Net

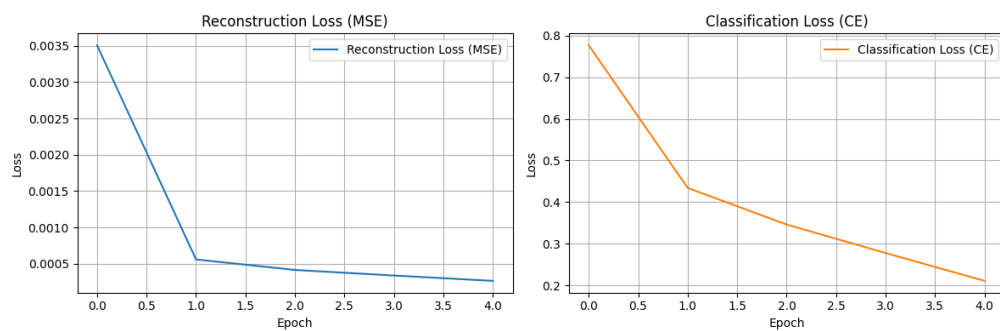


Figure 4: Loss Plots for Encoder-Decoder Y-Net

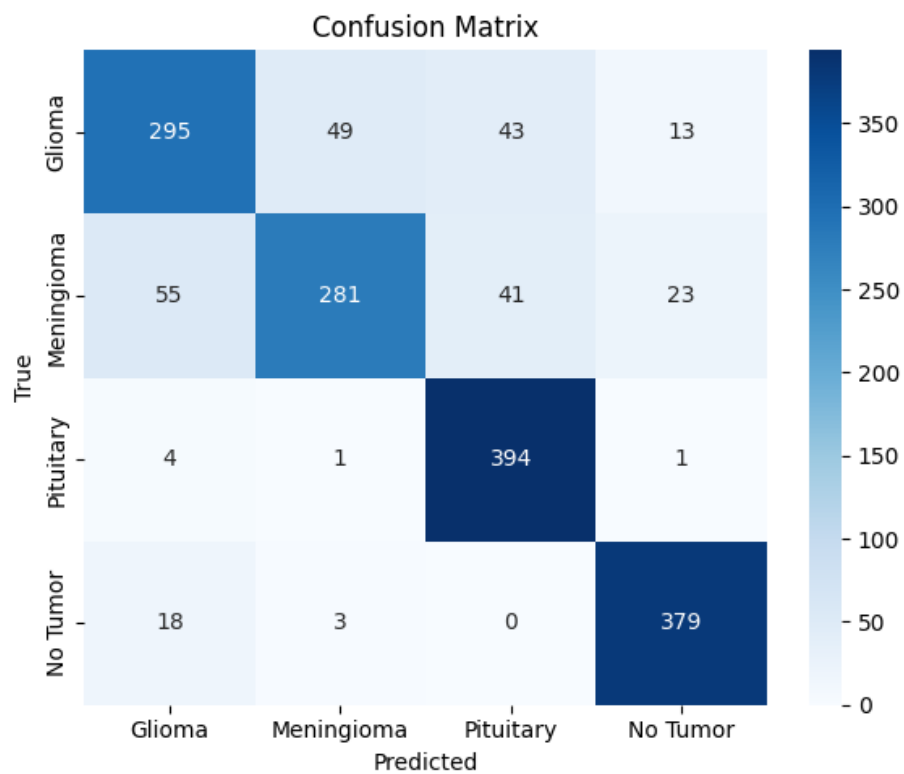


Figure 5: Confusion Matrix for Encoder only Y-Net

5 Conclusion

In this work, we presented a hybrid quantum-classical architecture for multi-class brain tumor classification, integrating the Quantum Feature Extractor (QuFeX) module into a Y-Net framework at the architectural bottleneck. Our experimental evaluation across four bottleneck configurations — QuFeX, classical U-Net, QCNN, and Quanvolution — under two architectural settings demonstrates the viability and competitive advantage of quantum-augmented feature extraction for medical image classification.

The results reveal a clear performance hierarchy. In the encoder-only setting, the classical U-Net bottleneck marginally outperformed QuFeX (80.64% vs. 79.11%), while both substantially surpassed the purely quantum alternatives, QCNN (76.95%) and Quanvolution (78.08%). However, when the full encoder-decoder architecture with skip connections was employed, QuFeX achieved the highest overall accuracy of 84.31%, surpassing the classical U-Net bottleneck (83.85%) and the quantum-only approaches by a considerable margin. This inversion highlights a critical finding: the representational benefit of QuFeX is most fully realized when the model is jointly optimized under a composite loss objective that combines classification and reconstruction, as the dual supervision compels the quantum bottleneck to preserve richer spatial and semantic information.

The confusion matrices further corroborate these findings. The full encoder-decoder QuFeX model demonstrated notably improved discrimination for the Glioma and No Tumor classes, reducing misclassification rates compared to the encoder-only variant. The Pituitary class remained consistently well-classified across all configurations, suggesting it presents more separable features in the MRI domain. These results collectively establish that embedding QuFeX within a complete encoder-decoder pipeline, guided by a weighted composite loss, represents the most effective configuration explored in this study, and motivates further investigation into deeper quantum circuit designs and larger-scale clinical datasets. Code available at: <https://github.com/bbhanja1809/Design-Lab-2026>

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